

Practical No. 9

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user ie. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code

```
/c/Users/john/OneDrive/Desktop/OSY_CD25D005
GNU nano 8.7
#include <stdio.h>

int main() {
    int pages[50], frames[10];
    int n, framesCount = 3;
    int i, j, k, pos, farthest;
    int faults = 0, hits = 0;
    int found;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &pages[i]);
    }

    for(i = 0; i < framesCount; i++)
        frames[i] = -1;

    for(i = 0; i < n; i++) {
        found = 0;

        // Check hit
        for(j = 0; j < framesCount; j++) {
            if(frames[j] == pages[i]) {
                hits++;
                found = 1;
                break;
            }
        }

        // If page fault
        if(found == 0) {
            pos = -1;
            farthest = i + 1;

            for(j = 0; j < framesCount; j++) {
                int nextUse = -1;

                for(k = i + 1; k < n; k++) {
                    if(frames[j] == pages[k]) {
                        nextUse = k;
                        break;
                    }
                }

                if(nextUse == -1) {
                    pos = j;
                    break;
                }
            }
        }
    }
}
```

```

/c/Users/john/OneDrive/Desktop/OSY_CD25D005
GNU nano 8.7
}

// If page fault
if(found == 0) {

    pos = -1;
    farthest = i + 1;

    for(j = 0; j < framesCount; j++) {

        int nextUse = -1;

        for(k = i + 1; k < n; k++) {
            if(frames[j] == pages[k]) {
                nextUse = k;
                break;
            }
        }

        if(nextUse == -1) {
            pos = j;
            break;
        }

        if(nextUse > farthest) {
            farthest = nextUse;
            pos = j;
        }
    }

    if(pos == -1)
        pos = 0;

    frames[pos] = pages[i];
    faults++;
}

printf("Frame status: ");
for(j = 0; j < framesCount; j++)
    printf("%d ", frames[j]);
printf("\n");
}

float hitPercent = ((float)hits / n) * 100;
float faultPercent = ((float)faults / n) * 100;

printf("\nTotal Page Hits = %d\n", hits);
printf("Total Page Faults = %d\n", faults);
printf("Hit Percentage = %.2f%%\n", hitPercent);
printf("Fault Percentage = %.2f%%\n", faultPercent);

return 0;
}

```

Output

```

/c/Users/john/OneDrive/Desktop/OSY_CD25D005

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ nano opr.c

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ gcc opr.c -o opr

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ ./opr
Enter number of pages: 12
Enter page reference string:
1 2 3 4 1 2 5 1 2 3 4 5
Frame status: 1 -1 -1
Frame status: 1 2 -1
Frame status: 1 2 3
Frame status: 1 2 4
Frame status: 1 2 4
Frame status: 1 2 4
Frame status: 1 2 5
Frame status: 1 2 5
Frame status: 1 2 5
Frame status: 3 2 5
Frame status: 4 2 5
Frame status: 4 2 5

Total Page Hits = 5
Total Page Faults = 7
Hit Percentage = 41.67%
Fault Percentage = 58.33%

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$

```